QPCODE: 26900066

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION OCTOBER 2026

FIFTH SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBOT300, MG5DSCBBT300 - ANGIOSPERM SYSTEMATICS AND ECONOMIC BOTANY

(2024 ADMISSION ONWARDS)

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt atleast one question from each course outcome to enhance their overall Outcome attainability

Duration: 1 Hour 30 minutes | Max. Marks: 50

Part A Short Answers

Answer any 10 out of 12 Questions, Each Question carries 1 mark

1	**Differentiate pinnately and palmately compound leaves.**	[U] / [CO3]
	Pinnately Compound: Leaflets are attached along a central axis called a rachis like a feather. Palmately Compound: Leaflets are attached at a single common point at the tip of the petiole, resembling fingers on a palm	
2	**What are the key features of a simple raceme?**	[K] / [CO3]
	Unbranched indeterminate main axis (peduncle) where pedicellate flowers produced in acropetal succession.(e.g., Crotalaria)	
3	**What type of placentation is present in pea?**	[K] / [CO3]
	Marginal placentation	
4	**Name the genus which produces hypanthodium inflorescence.**	[K] / [CO3]
	Ficus (Mark can be given for 'Fig plant' also)	
5	**What is the main purpose of a taxonomic key?**	[U] / [CO1]
	To facilitate the identification of unknown plant specimens based on a series of contrasting choices (dichotomous couplets and leads).	
6	**Which characteristic climbing structure is commonly found in Cucurbitaceae?**	[K] / [CO3]
	Tendrils	
7	**What is the basis of Linnaeus' classification?**	[K] / [CO2]
	Linneaus' classification is an example of artificial system of classification which was based on number, position, cohesion and adhesion of reproductive parts.	

8	**What are the pollen masses of orchids called?**		[U] / [CO3]
		Pollinia	
9	**Which type of aestivation is characteristic of corolla in Fabaceae?**		[K] / [CO3]
		Vexillary/ Descending Imbricate aestivation.	
10	**Mention one limitation of Bentham and Hooker's system.**		[K] / [CO2]
		Insertion of Gymnosperms between Dicotyledons and Monocotyledons/Advanced families placed before primitive families in certain groups (Any other relevant point)	
11	**Give the botanical name of banana.**		
		Musa X paradisiaca (*Musa acuminata*, *Musa balbisiana*)	
12	**Give the botanical name of foxtail millet.**		
		Setaria italica	
			[10x1 = 10]

Part B -Short Essay Type Questions
Answer any 6 out of 8 Questions, Each Question carries 5 marks

13	**Explain the symmetry of flowers with suitable examples.**		[K] / [CO3]
		Flower symmetry describes how the petals, sepals, and other floral parts can be divided into identical halves. Three main types flowers are there based on their symmetry. Actinomorphic (Radial Symmetry): A flower that can be divided into two equal/identical halves along any vertical radial plane passing through the center. e.g.*Hibiscus*, *Allamanda*, *Ixora*, etc Zygomorphic (Bilateral Symmetry): A flower that can be divided into two identical halves along only one single vertical plane passing through the center. e.g. *Clitoria, Crotalaria,* Orchids, etc. Asymmetric: A flower that cannot be cut into two equal halves along any vertical plane through the center. e.g. *Canna indica*.	
14	**What are simple dry fruits? Explain its different types with suitable examples.**		[K] / [CO3]
		They are simple fruits formed from a single monocarpellary or syncarpous pistil where the pericarp becomes dry and non-fleshy at maturity. Types of Simple Dry Fruits: Dry dehiscent , dry indehiscent and schizocarpic fruits 1) Dry Dehiscent fruits- Multiseeded fruits whose pericarp splits open spontaneously at maturity to release seeds. ● Legume (Pod): Splits along both dorsal and ventral sutures (e.g., Pea, Bean) ● Follicle: Splits along one suture only (e.g., Calotropis) ● Capsule: Multi-locular fruit dehiscing in various ways (e.g., Datura, Cotton) 2) Dry Indehiscent Fruits: Single seeded fruits whose pericarp does not split open at maturity; seed released when pericarp decays. ● Caryopsis: Seed coat fused completely with pericarp (e.g., Rice, Wheat). ● Cypsela: Derived from inferior bicarpellary ovary with persistent pappus (e.g., Sunflower). ● Nut: Hard, woody pericarp (e.g., Cashew nut, Hazelnut). 3) Schizocarpic Fruits: Intermediate type; breaks up at maturity into single-seeded units (mericarps or cocci). ● Lomentum: Constricted between seeds (e.g., Mimosa pudica). ● Cremocarp: Splitting into two mericarps attached to a carpophore (e.g., Coriander).	

15	**Comment on the importance of floras, revisions and monographs in plant taxonomy.**		[U] / [CO5]
		Floras: Books containing comprehensive descriptions, taxonomic keys, and distribution details of all plant species inhabiting a specific geographical area (e.g., Flora of British India by J.D. Hooker, Flora of Presidency of Madras by J.S. Gamble). Importance of Floras: Essential field guide for rapid identification, local conservation strategies, biodiversity assessment, and natural resource documentation. **Monographs:** Exhaustive, global taxonomic accounts of a single biological taxon (usually a genus or family) including all species, synonomy, morphology, phylogeny, and geographical distribution worldwide. Importance of monographs: Serves as the ultimate authority on worldwide variation within a specific family/genus; solves taxonomic ambiguities. **Revisions:** Updated taxonomic re-examinations of previously published monographs or floristic treatments of specific taxa within a region. Importance of revisions: Incorporates new research findings (molecular, anatomical, palynological), corrects nomenclature error, and updates classification in light of contemporary data.	
16	**Identify the diagnostic features of Acanthaceae that are useful for family identification**.		[U] / [CO3]
		Vegetative Characters: ● Herbaceous or shrubby habit with opposite decussate leaves. ● Stems frequently swollen at nodes; cystoliths (calcium carbonate deposits) present in epidermal cells. **Floral Characters:** ● Prominent, often brightly colored, overlapping bracts and bracteoles subtending flowers. ● Corolla: Bilabiate (2-lipped) or sub-equally 5-lobed, zygomorphic. ● Androecium: Usually 2 or 4 (didynamous) stamens inserted on the corolla tube. ● Gynoecium: Bicarpellary, syncarpous, superior ovary with axile placentation. ● Fruit is a loculicidal capsule. ● Seeds mounted on hook-like outgrowth of funicle called retinaculum (ejaculater) which helps in seed dispersal. (**Any five relevant points can be considered for full marks**)	

17	**Write a short note on ICN.**	[K] / [CO1]

ICN (International Code of Nomenclature for algae, fungi, and plants), formerly known as ICBN (renamed at the Melbourne Congress in 2011). The International Code of Nomenclature for algae, fungi, and plants is the set of rules and recommendations dealing with the formal botanical names that are given to plants, fungi and a few other groups of organisms, all those "traditionally treated as algae, fungi, or plants".

Key Principles:

- Botanical nomenclature is independent of zoological and bacteriological nomenclature.
- Type Concept (Nomenclatural Types): Application of names of taxonomic groups is determined by means of nomenclatural types (holotype, isotype, lectotype, neotype).
- Principle of Priority: The correct name of a taxon is the earliest validly published name in accordance with the rules
- Unique Name Rule: Each taxonomic group with a particular circumscription, position, and rank can bear only one correct name.
- Scientific names are treated as Latin regardless of derivation.
- Rules of nomenclature are retroactive unless expressly limited.

(Any five valid points about ICN can be considered for full marks)

18	**Analyse the advanced features of the family Asteraceae that make it one of the most successful angiosperm families.**	[An] / [CO3]

- Inflorescence Aggregation (Capitulum/Head): Hundreds of small flowers (florets) gathered together into a single head mimicking one large flower, maximizing pollinator visibility with minimal tissue expenditure.
- Specialized Pollination Mechanism: Piston/plunger (syngenesious) pollination mechanism ensuring efficient cross-pollination and backup self-pollination.
- Modification of Calyx into Pappus: Calyx modified into hairy, feathery, or awned structures (pappus) aiding in widespread anemochorous (wind) seed dispersal.
- The syngenesious condition—where anthers are fused laterally to form a hollow tube while filaments remain free—plays a crucial role in the evolutionary and reproductive success of the Asteraceae family
- Epigynous Flower & Single Seed Protection: Inferior bicarpellary ovary yielding a single-seeded cypsela fruit, directing energy into a single viable offspring per floret.
- Chemical Defense Mechanisms: Rich production of secondary metabolites discouraging herbivores and pathogens.
- Ecological Adaptability: Herbaceous habit with short life cycles enabling rapid colonization across diverse biomes.

(*Any five valid points can be considered for full marks*)

19 **Compare the vegetative and floral characters of Annonaceae and Malvaceae**. [A] / [CO3]

Character	**Annonaceae**	**Malvaceae**
Habit & Secretions	Woody trees/shrubs, aromatic, lacking mucilage	Herbs, shrubs, or trees with abundant mucilage glands
Leaves	Alternate, simple, exstipulate, entire margin	Alternate, simple, stipulate, palmately veined/lobed
Epicalyx	Absent	Present (whorl of bracteoles below calyx)
Calyx & Corolla	Usually 3 sepals and 6 petals (2 whorls of 3)	5 sepals (valvate), 5 petals (twisted/contorted)
Stamens	Numerous, spirally arranged on spirally elongated torus, short filaments, prominent connective tissue (Hooded anthers)	Indefinite, monadelphous (united into a staminal tube), 1-celled/monothecous anthers
Gynoecium	Polycarpellary, apocarpous, superior	Syncarpous (5 to many carpels), superior, axile placentation
Fruit	Aggregate fruit (Etaerio of berries)	Loculicidal capsule or schizocarp
(**Any five valid differences can be considered for full marks**)		

20 **Compare tobacco and neem based on their botanical names, families and insecticidal parts.** [U] / [CO2]

<u>TOBACCO</u>
Botanical Name: *Nicotiana tabacum*
Family: Solanaceae
Insecticidal part : Leaves (active compound: Nicotine)

<u>NEEM</u>
Botanical Name: *Azadirachta indica*
Family: Meliaceae
Insecticidal part : Seeds / Seed Oil & Leaves (active compound: Azadirachtin)

[6x5 = 30]

Part C- Essay Type Questions
Answer any 1 out of 2, Each Question carries 10 marks

21	**What are the main features of cymose type inflorescences? Explain each type of cymose inflorescence with suitable examples.**		[U] / [CO3]
		■ Main Features of Cymose Inflorescence: 1) Determinate Growth of peduncle: Main axis (peduncle) ends in a flower, stopping further apical elongation. Further growth by forming lateral branches.Lateral branches develop beneath the terminal flower and continue growth in a similar manner. 2) Basipetal /Centrifugal succession of flowers: Flowers open in a basipetal (top flower opens first,lower flowers open later) or centrifugal (center flower opens first, lateral flowers open later) succession. ■ Types of Cymose Inflorescence: 1) **Uniparous Cyme (Monochasial Cyme)**: Main axis ends in a flower; produces only one lateral branch at a time ending in a flower. -Subtypes of monochasial cyme: A) Helicoid Cyme: Lateral branches develop successively on the same side, forming a spiral (e.g., *Hamelia*). B) Scorpioid Cyme: Lateral branches develop alternately on opposite sides in a zig-zag manner (e.g., *Heliotropium* **2) Simple Cyme** ❖ Consists of three flowers. ❖ Peduncle terminates in a flower which opens first. ❖ Two younger flowers of same age arises from the node just below. (e.g., *Jasminum)* **3) Biparous Cyme (Dichasial Cyme):** ❖ Peduncle terminates in a flower ❖ It develops two lateral branches that develop into a simple cyme. (e.g., *Clerodendrum*) **4) Multiparous cyme (Polychasial Cyme)** ❖ Main axis terminates in a flower and produces more than two lateral branches in a whorl from its base, each ending in a flower. (e.g., *Calotropis)*	
22	**Define herbarium. Explain the steps in herbarium technique in detail.**		[U] / [CO4]
		A herbarium is a curated collection of dried, pressed, mounted plant specimens arranged systematically according to an accepted system of classification, serving as a permanent reference source for taxonomic research.	

Steps in Herbarium Technique:

1) Field Collection:

- Collect healthy plant specimens containing flowers, fruits, and vegetative organs.
- Collect field data (habitat, locality, date, plant altitude, flower color, collector name) in a field notebook.
- Store samples in a Vasculum or polythene bag to prevent moisture loss.

2) Pressing and Drying :

- Spread plant parts carefully without overlapping on blotting sheets or newspapers.
- Place sheets inside a heavy plant press.
- Change blotting paper regularly (every 24–48 hours) to prevent fungal decay and discoloration.
- Complete drying can be accelerated using warm air driers.

3) Poisoning / Chemical Treatment:

- Treat dry specimens with chemical preservatives (Mercuric chloride or Lauryl pentachlorophenate (LPCP) solution) or deep freezing to protect against insect pests.

4) Mounting and Labelling:

- Mount the dried specimen onto standard heavy herbarium sheets ((approximately 16.5 inches × 11.5 inches or 42 cm × 29 cm) using glue or adhesive tapes.
- Attach a standard printed herbarium label at the bottom right corner of the sheet.
- Label should include Name of Institution, Botanical Name, Family, Vernacular Name, Locality, Date of Collection, Ecological Habitat, and Collector's Name.

5) Filing and Preservation:

- Store sheets systematically inside wooden/metallic herbarium cabinets arranged by standard classification systems (e.g., Bentham & Hooker).
- Use naphthalene balls or insecticides inside storage cabinets periodically.

[1x10 = 10]